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MILITARY SPECIFICATION

TRUCKS, TANK: SEWAGE AND WASTE, 2,000 GALLON, DIESEL AND GASOLINE ENGINE DRIVEN, 32,000 AND 36,000 GVW, 4x4 and 6x6, COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven, four-wheel, four-wheel drive and six-wheel, six-wheel drive vehicles having minimum gross vehicle weights (GVW) of 32,000 and 36,000 pounds, for normal operating conditions. Vehicles are commercial trucks equipped with a 2,000 gallon tank and pump for collecting and discharging sewage and waste. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The vehicle shall be one of the following classes, as specified (see 6.2):

Vehicle class	Minimum GVW rating (pounds)
A	32,000
B	36,000.

1.2.1 Model. Model of vehicle shall be determined by the number of wheels and the number of driving wheels as follows:

Model 4x4	Four wheels, four wheels driving
Model 6x6	Six wheels, six wheels driving.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Specification Comment Sheet appearing at the end of this document or by letter.

AMSC N/A

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting, and Starting.
*	ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).
	ZZ-H-451	- Hose, Fire, Woven-Jacketed, Rubber- Or Cambric-Lined, With Couplings.
	ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water.

STANDARDS
* FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

*	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
	MS51118	- Pintle Assembly, Towing - 40,000 lbs. Capacity, Manual Release.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.
Regulations Governing Transportation of Dangerous Articles In Tank Motor Vehicles.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations 40 CFR and the Federal Register and should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards
ASTM Designation: A120 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses (DoD adopted).

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* SAE, INC.
SAE Standards and Recommended Practices.

	J245	- Engine Rating Code - Spark Ignition.
*	J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors and Trailers (DoD adopted).
	J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
	J537	- Storage Batteries (DoD adopted).

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	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
	J683	- Tire Chain Clearance - Trucks, Buses, and Combinations of Vehicles (DoD adopted).
	J688	- Truck Ability Prediction Procedure (DoD adopted).
	J704	- Openings for Six and Eight Bolt Trucks Transmission Mounted power Take-Offs (DoD adopted).
	J706	- Rating of Winches.
	J816	- Engine Test Code - Spark Ignition and Diesel.
	J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
	J1067	- Seven-Conductor Jacketed Cable for Truck and Trailer Connectors.
*	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to the SAE, INC., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Incorporated, 3200 W. Market Street, Akron, OH 44313.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.10, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented and rated in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. Treatment, painting, markings, and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2).

3.1.1.1.1 Instruction plate. A photoetched corrosion-resistant plate showing a schematic of all valves, valve controls, equipment, and complete instructions for operating body related components shall be fastened on the inside of the rear cabinet door.

* 3.1.1.2 Rustproofing. When specified (see 6.2), the truck cab, chassis and chassis underside shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2) tropical rustproofing, in accordance with FED-STD-297, shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in manual transmissions, transfer case and rear axle(s) shall be of the permanent magnet type.

3.1.1.4 Heavy duty cooling system. A heavy duty cooling system shall be furnished that shall maintain engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

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3.1.1.5 Coolant system and indicators. Coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than 2 quart capacity. A high coolant temperature or low coolant level alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

3.1.1.6 Towing devices. Not less than two hooks, loops, pins, or a single center mounted towing eye, for towing the vehicle shall be furnished on the front of the vehicle. When specified (see 6.2), not less than two hooks, loops, or pins for towing the vehicle shall be furnished on the rear of the vehicle.

* 3.1.1.7 Trailer towing package. A trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle, a trailer brake control system, and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS-51118. The pintle shall be installed on the chassis frame with reinforcements to transfer a tongue load of not less than 4,000 pounds and a drawbar load of not less than 40,000 pounds directly to the chassis rails. The rearmost portion of the pintle shall be forward but not more than 4 inches forward of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the vehicle furnished. The attachment devices shall be capable of accommodating a standard grab hook (4-9/16 inches wide, 1-3/16 inches thick, 25/32-inch throat width) for a 5/8-inch chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle. The trailer brake control system shall conform to 3.4.11.2.

* 3.1.1.8 Rear wheel splash and stone throw protection. Rear wheels shall be provided with rubber mud flaps at the rear of the rear wheels, installed in accordance with SAE J682. A metal strip, not less than 1/8 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent bolt heads or bolt nuts from damaging the mud flap.

3.1.1.9 Radiator shutters. When specified (see 6.2), radiator shutters shall be furnished. The radiator shutters shall be thermostatically controlled, air actuated and of the multi-vane type. The radiator shutters shall maintain the correct coolant temperature. Means shall be provided for manual operation. The shutters shall remain in the open position should the thermostatic controls fail.

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* 3.1.1.10 Brushguard. When specified (see 6.2), the vehicle shall be equipped with a radiator and headlamp brushguard. When the headlamps are recessed in and protected by the front bumper, a headlamp brushguard is not required.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to the EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories, and equipment; body; and a full complement of fuel, lubricants, and coolant.

3.2.5 Payload. Payload shall consist of sewage, calculated at 9 pounds per gallon, and shall be not less than 18,000 pounds.

3.2.6 Gross vehicle weight. The minimum gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 175 pounds) and a payload distributed as specified in 3.2.7, to provide not less than the minimum GVW specified in 1.2. In addition, the actual GVW of the vehicle furnished shall exceed the minimum GVW specified in 1.2 when required to provide a payload not less than that specified in 3.2.5.

3.2.7 Weight distribution. The distribution of the GVW for the purpose of establishing suspension, axle and tire capacities shall be determined with the payload uniformly distributed over the vehicle body load area.

3.2.8 Ratings. Vehicle and component ratings shall be the manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be submitted to the engineering office of the procuring activity.

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* 3.2.9 Overall Width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires, shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

* 3.2.10 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds. High and low speed requirements shall be met with the vehicle loaded to required GVW. Performance requirements shall be met with the front wheel drive disengaged, except that when equipped with inter-axle compensating devices, vehicle shall meet the performance requirements with front wheel drive engaged.

3.3.1.1 High speed gradeability. The vehicle shall ascend a 0.50 percent grade at 50 miles per hour (mph). Gradeability requirements shall be met with the transmission in direct drive and, when an auxiliary transmission is furnished, with the auxiliary transmission in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of recommended governed speed, and shall provide a vehicle speed of not more than 3 mph.

* 3.3.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 55 mph. Conformance to geared speed specified shall be determined in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{governed speed (rpm)}}{\text{total gear reduction x tire factor (see 6.3)}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its required GVW on a 30 percent grade. The service brakes shall stop the vehicle, loaded to required GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

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* 3.4.1.1 Diesel engine driven. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, internal combustion, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

* 3.4.1.2 Gasoline engine driven. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall provide the required vehicle performance when operated on unleaded fuel with a research octane rating of 91 at engine speed not more than manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel when used in accordance with the operator's manual. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J245 or SAE J1349. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to maximum recommended operating speed.

3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.6 Power plant heaters and fuel warmer. When specified (see 6.2), coolant heater, engine oil heater, and a fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block to a single three pronged (male), weatherproof, slave receptable for receiving external power and grounding vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall

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include a matching female connector at the vehicle end and a standard three pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c) (7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. Engine coolant thermostat with a operating range of 170°F to 195°F shall be installed.

(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening.

(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

(d) For the auxiliary gasoline engine (see 3.5.5.2), a 300-watt immersion type heater with 170°F to 195°F thermostat shall be installed in oil pan through any convenient opening.

(e) A fuel warmer or pre-heater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical systems. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12 or 24-volt direct current (dc) starting system with 12-volt dc lighting system, and with not less than a 60-ampere alternator, shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type

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ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.2 Ignition system. For gasoline engine driven vehicles, an ignition system of 12-volt dc potential shall be furnished. An alternator of not less than 40 amperes rated capacity and which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished.

* 3.4.2.3 Lighting. All vehicle lights, reflectors, and wiring shall be in conformance with Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails or mounted on vehicle bumpers unless recessed or fully protected. Unused headlight cavities shall be covered flush with the adjoining body surfaces in a neat and workmanlike manner.

3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589 class A and shall be mounted on the steering column. Vehicle shall be provided with double faced front signal units and with single faced rear signal units installed in accordance with SAE J588 except that turn signals shall not be mounted on engine compartment hood. Turn signal operating units shall have visible and audible type flash indicators.

3.4.2.5 Trailer lighting cable. When specified (see 6.2), a trailer lighting cable, conforming to SAE J1067, shall be furnished. The cable shall be coiled and shall be not less than 110 inches long when fully extended. Both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

* 3.4.2.6 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity ratings, and the total cold cranking ratings at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table I. The batteries shall be of the maintenance free type having the maintenance-free characteristics listed in W-B-131.

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Table I. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel (less than 220 gross horsepower)	320	1200
Diesel (220 gross horsepower and over)	480	1800
Gasoline	100	450

* 3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil-bath type air cleaner, oil capacity shall be not less than 1 quart.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) shall be of not less than 90 gallon total capacity. When more than one tank is furnished on a diesel engine driven vehicle, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on a gasoline engine driven vehicle, a selector valve connecting either tank to engine fuel intake shall be provided, and means shall be provided to monitor the fuel level of either tank from a single fuel gage.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. Vertical exhaust mufflers, if furnished and capable of being reached easily by personnel entering or leaving either side of the cab, shall be provided with a heat shield. Vertical exhaust systems, if furnished, shall be provided with a hinged rain cap.

* 3.4.4.1 Spark arrester. A spark arrester shall be furnished, except on vehicles with turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

* 3.4.5 Transmission. Unless otherwise specified (see 3.4.5.3), the vehicle shall be furnished with a manually shifted transmission. Input torque capacity of the transmission shall be at least equal to the

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maximum torque delivered by the engine. The vehicle shall be provided with not less than 10 forward speeds. Gear ratios in the transmission and axles shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide required vehicle performance (see 3.3.1). The transmission shall be provided with power takeoff opening(s) conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds.

* 3.4.5.1 Transfer case. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between the front and rear axles, the transfer case shall provide for driver selection of either two-wheel, four-wheel or six-wheel drive as applicable. When furnished, interaxle compensating devices shall provide for positive transfer of power to all driving axles. A one- or two-speed transfer case shall be furnished. When specified (see 6.2), a two-speed transfer case shall be furnished.

3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the vehicle and engine furnished, with clutch torque capacity exceeding maximum delivered engine torque.

* 3.4.5.3 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with automatic or semiautomatic transmission. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission furnished shall provide continuous drive. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The driving range selector shall provide a normal drive position and a second low speed, high power position. The transmission shall be provided with power takeoff opening(s), conforming to SAE J704.

* 3.4.5.4 Power takeoff. When a power takeoff is furnished, it shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction. The driveshaft shall be protected in a manner similar to that specified for buses by Federal Motor Carrier Safety Regulation 393.89.

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3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the vehicle furnished. Model 4x4 vehicle shall have frame with reinforcements extending at least from the rear of the front-suspension, rear hanger bracket to the front of the rear-spring, front hanger bracket. Model 6x6 vehicle shall have frame with reinforcements extending at least from the rear of the front-suspension rear hanger bracket to the bogie trunnion mounting bracket. Reinforcements shall provide sufficient structural strength in the chassis frame through increased bending moment to at least equal the loads imposed with the body loaded to provide the specified GVW. Chassis frame rails shall not project beyond the rear end of the tank body or rear bumper.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with vehicle loaded to required GVW. When spring capacity is rated at the spring pads, unsprung weight shall be deducted. Vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to required GVW. Single reduction front and rear axles shall be furnished.

3.4.9.1 Rear bogie. The rear bogie on model 6x6 vehicle shall be of the four-wheel type, complete with axles, springs, torque rods and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles, and a manually or automatically controlled lockout assuring equal power to each rear axle. The manual lockout control shall be located in the truck cab. Axle gear ratios shall provide performance specified in 3.3.1.

3.4.9.2 Traction control. When specified (see 6.2), a traction control shall be furnished on the rear axle of model 4x4 vehicles, and on either drive axle of the rear bogie of model 6x6 vehicles. The traction control shall actuate automatically to insure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Rim and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating)

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shall be the same for all tires on the vehicle. When specified (see 6.2), the vehicle shall be equipped with wide base type tires and wheels for the front and rear axles in lieu of dual wheels and tires. When wide base tires are furnished, the wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall be tube or tubeless type with nondirectional mud and snow tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the vehicle loaded to required GVW. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381, with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Inner tubes. For tube tires, inner tubes shall be of heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of, or plated with, corrosion-resistant material. The carrier design shall enable removal or mounting of spare wheel assembly using only tools specified in 3.4.16.1.

* 3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

* 3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the front axle of the vehicle.

* 3.4.10.6 Tire chain clearance. Tire chain clearance, in accordance with SAE J683, shall be furnished. Allowance for spring deflection shall be included.

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3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b), 393.43 and 393.45 through 393.52.

* 3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes. Brakes shall be furnished on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm);
(b) Air storage reservoir(s) with not less than 2,000 cubic inches total capacity; each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank;
(c) Foot control, suspended or treadle type;
(d) Air control valves;
(e) Air pressure gage, visible to the driver;
(f) Low-air-pressure warning, visible and audible;
(g) Service brake stop lamp switch;
(h) Alcohol aspirator with unbreakable container;
(i) Automatic moisture ejector;
* (j) When specified (see 6.2), in lieu of subparagraph (h), an air dryer shall be furnished. The air dryer shall be of either the desiccant or after-cooled type.

3.4.11.2 Trailer brake control system. In addition to components specified in 3.4.11.1, a trailer brake control system shall be furnished. The trailer brake control system shall include:

(a) Identification of emergency and service lines;
(b) Coincident control of trailer brakes with prime mover foot control;
(c) Independent hand control for trailer brakes;
(d) Prime mover protection valve with dash control and automatic break-away feature;
(e) Trailer stoplight operative with foot brake and with hand control for trailer brakes;
* (f) Two SAE J844 coiled air hoses, not less than 110 inches long with SAE J318 gladhand couplers on both ends of hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment;
* (g) Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to preclude interference with trailer;
(h) Dummy gladhand couplers with chains.

3.4.11.3 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for

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controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. The chassis manufacturer shall furnish any type of his standard or optional full width cab. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with external, key-operated lock. Drip rails shall be installed above the cab doors. Unless otherwise specified, the cab shall have upholstered, full width, adjustable seat and back, or individual, adjustable driver's seat and individual passenger seat. When specified (see 6.2), an individual adjustable driver's seat and an individual passenger seat shall be provided. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Safety grips shall be provided on each side of the cab to assist personnel in entering and leaving the cab. Interior lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be installed on the vehicle. When step height exceeds 24 inches, a secondary step shall be provided.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be of the multispeed type and operated by air or electric motor(s).

3.4.15 Bumpers. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided, except when a front mounted winch is specified (see 3.4.22).

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86.

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* 3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, anti-skid chains, warning flares and reflectors shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly and shall include at least a hydraulic jack, jack handle and wheel-nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to driver when seated in driving position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Dual sunvisiors
(h) Driver's compartment ventilator other than window
* (i) Tachometer.

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3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas inclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 20 square inches of reflective area. The mirrors shall have not less than three supporting arms.

3.4.21 Horn. Manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be provided.

* 3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel, or in the engine compartment in a readable location.

3.4.23 Back-up alarm. When specified (see 6.2), the vehicle shall be provided with an audible, pulsating signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

* 3.4.24 Air conditioning. When specified (see 6.2), the vehicle shall be equipped with the manufacturer's standard air conditioning system. The air conditioning system shall include tinted windshield and tinted glass.

* 3.4.25 Front mounted winch. When specified (see 6.2), a winch assembly, mounted on the front of the vehicle and powered by a power takeoff, or hydraulically operated, shall be provided. The winch shall conform to SAE J706. The winch shall be single drum type and shall have one forward and one reverse speed. All winch controls shall be mounted inside the cab. Winch controls shall be located to provide no interference with the entrance or exit of the driver. The winch shall have a rated single-line pull capacity of not less than 20,000 pounds pull on the bare drum. The winch line speed on the bare drum shall be between 25 and 30 feet per minute at an engine speed of 1,200 rpm for manual transmission and 2,000 rpm for automatic or semiautomatic transmissions. The winch shall be wound with not less than 185 feet of 5/8-inch diameter, preformed 6x37, improved plow steel, wire strand core wire rope and shall be equipped with end chain and hook. An integral, adjustable, automatic safety brake shall be provided. The winch shall be equipped with a roller guide. Angles of the winch drive line U-joints shall be not greater than 16-1/2 degrees.

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3.4.25.1 Winch drum guard. A winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges. The guard shall consist of not less than 1/4-inch vertical side plates, conforming to the outside radius of the drum flanges. Six bars, 3/8 inch by 1-1/4 inches, shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the drum flanges shall be not greater than one half the specified cable diameter.

3.4.25.2 Front bumper. When a front mounted winch is specified the bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard (see 3.4.22.3). The bumper shall be either a channel (see 3.4.15) or a pipe type front bumper. When a pipe type front bumper is furnished, the nominal diameter shall be not less than three inches, and the wall thickness shall be not less than that specified in Schedule 40 of ASTM A120. The pipe type front bumper shall have half-round ball ends.

3.4.25.3 Combination step plate and gravel guard. When a front mounted winch is specified, a combination step plate and gravel guard shall be furnished. The step plate shall be fabricated of not less than 14 gage (0.0747 inch) steel tread plate exclusive of projections. The step plate shall be secured to the front bumper and shall be not less than 1/4 inch and not more than 3/8 inch from the chassis sheet metal. The step plate shall be capable of supporting 300 pounds per square foot. The step plate shall not deflect more than 1/8 inch under the loads imposed.

3.5 Tank design. The vehicle shall be equipped with a 2,000 gallon capacity tank for collecting and discharging sewage and waste water. The tank shall be mounted on a chassis with a rear-of-cab to centerline of rear axle dimension of not less than 120 inches. The pumping system shall have a design capacity of not less than 300 gallons per minute (gpm) under a total dynamic head of 45 feet through a 3-inch smooth bore rubber hose. The tank body shall be round, elliptical or modified rectangular section. The length of the tank shall be 162 inches plus or minus 6 inches. The collection of sewage and waste water shall be by vacuum into the pump and discharge into the tank by pressure. Discharge of tank contents shall be by the pump or by gravity, at the operator's option. Underslung cabinets forward of the rear axle and a rear cabinet for the pump and auxiliary engine shall be provided. The tank shall be designed and constructed in accordance with DoT Safety Regulation 178.341, Specification MC306.

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3.5.1 Tank construction. The tank shall be of welded construction using type 304 stainless steel. The tank interior shall have a 2B finish. The tank shall be smooth to preclude any areas retaining water when the tank is drained. The tank shall have a single compartment. The tank body may be provided with a slope for removal of liquids. When a slope is provided, it shall not exceed one percent of the tank body length.

3.5.1.1 Tank support. Full length box type, V type or Z type longitudinal subframe members shall be furnished. Full length welding pads of not less than 12 gage (0.1046 inch) type 304 stainless steel shall extend not less than 1 inch beyond the flanges on the longitudinal subframe members and shall be continuously welded to the tank proper. Box type longitudinal subframe members shall be fabricated with not less than a 7 gage (0.1793 inch) mild steel outer leg and bottom with not less than a 12 gage (0.1046 inch) mild steel inner leg. A flange not less than 1 inch wide shall be provided at the top of both the outer and inner legs. Flanges shall be continuously welded to the pad the full length of the tank. V type longitudinal subframe members shall be fabricated with not less than 7 gage (0.1793 inch) mild steel inner and outer legs. A flange not less than 1 inch wide shall be provided at the top of both the inner and outer legs. Flanges shall be continuously welded to the pad the full length of the tank. Z type longitudinal subframe members shall be fabricated entirely of not less than 7 gage (0.1793 inch) mild steel. The width of the upper Z member shall be not less than 2 inches and shall be continuously welded to the pad the full length of the tank along both sides of the Z. With the Z type construction, additional type 304 stainless steel pads shall be continuously welded to the tank proper at the front, rear, at each bulkhead and at each baffle. Bolsters of 3/16-inch mild steel shall be properly integrated with the Z type longitudinal and continuously welded to the pads at the front, at the rear, at each bulkhead and at each baffle.

* 3.5.1.2 Baffles. Baffles shall be provided. The linear distance between the baffles and the ends of the tank shall not exceed 60 inches. Each baffle shall have a top and bottom half round or full round opening with not less than a 3-3/8 inch radius, and a manway opening of not less than 18 inches in diameter. The 18-inch openings shall be staggered to one another to resist surging of the payload. Baffles shall be of type 304 stainless steel and provided with a 2B finish. Baffles shall be flanged, dished or reinforced. The tank interior shall be free of crevices that would entrap moisture.

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* 3.5.1.3 Manhole. The tank shall be equipped with a round manhole opening not less than 20 inches in diameter with a 10 inch fill opening (see 3.5.1.4). The manhole shall be located on the longitudinal centerline of the tank approximately midway between the front and rear of the tank. The manhole shall be provided with a non-ferrous hinged or bolted liquid-tight cover. The manhole cover shall have a watertight neoprene gasket, non-ferrous hinges and a non-ferrous latch.

3.5.1.4 Fill opening. A fill opening equipped with not less than a 10 inch diameter, hinged, liquid tight cover equipped with a neoprene gasket, and locking clamps shall be provided. The fill opening cover shall be hinged to the manhole cover. The fill opening shall be provided with a removable 6 inch diameter non-ferrous tube, 12 inches in length and flanged on one end to prevent loss of the tube inside the tank. The fill opening and cover shall be an integral part of the manhole cover and of the same material composition.

3.5.1.5 Vents. The vehicle shall be provided with self-acting pressure-vacuum vents to equalize the internal and external air pressure during loading and unloading operations when the manhole and fill opening are sealed. Vents shall preclude freezing and shall be designed so that the vents can be easily cleaned. A corrosion-resistant warning plate shall be furnished at the fill opening and the discharge opening stating that the vents must be operable during loading and unloading operations.

3.5.2 Catwalk and ladder. A catwalk over the top of the tank, not less than 72 inches in length shall be provided. The catwalk shall be of aluminum alloy or stainless steel grip-strut type grating. The catwalk shall be located on risers above the tank top. The catwalk shall be centrally located at the manhole. An access ladder or steps shall be provided for access to the catwalk. The ladder or steps shall be on the curbside of the vehicle, at the manhole.

3.5.3 Hose tubes. Hose tubes adequate to stow the suction hose specified in 3.6.6 shall be provided on each side of the tank skirting. The hose tube openings shall be in the pump compartment. The hose tubes shall be protected from weather and shall have provisions for draining and cleaning.

* 3.5.3.1 Skirting. Skirting down to not less than 15 inches below the top of the chassis frame rails shall be provided. Skirting shall be the full length of the body on both sides, across the rear of the vehicle to the inside of the chassis frame rails, and across the front

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of the tank to the chassis frame rails. Cutouts for the wheels shall be rounded. The radius of the rounded position of the cutout shall be approximately 1-1/4 times the tire radius. The skirting shall be constructed of not less than 16 gage (0.0598 inch) sheet steel. Clearance for tire chains shall be provided in accordance with SAE J683. The top surface of the skirting shall be reinforced and covered completely with abrasive type paint (ship's grip) for use as a catwalk. The skirting shall contain hose storage tubes and curbside and streetside cabinets.

3.5.4 Cabinets.

3.5.4.1 Pump cabinet (rear). A pump cabinet, having the same general sectional configuration as the tank, shall be furnished at the rear of the tank. The cabinet shall be of a full-flanged, flush type construction. The cabinet shall be of sufficient size to accommodate the pump, auxiliary engine, pump heater, 25 feet of 3-inch discharge hose, manifold, fittings, control panel and tools.

3.5.4.1.1 Pump cabinet construction. The cabinet shall be constructed of not less than 16 gage (0.0598 inch) sheet steel shall be stormproof and shall include provisions for flushing and draining. Latches or other devices to hold each door open at angles of not less than 90 degrees and 180 degrees shall be provided. Rubber door bumpers shall be provided. Doors shall be provided with neoprene rubber gaskets. Doors shall be furnished with cam type locks with vertical drawbolts at the door header and door sill. Door handles shall be capable of operation by personnel wearing arctic gloves. Door locks shall be capable of being locked with a 1-inch padlock. Hinges for the doors shall be continuous rod or hem type with stainless steel hinge pins and stainless steel, bronze, nylon or nylon insert sockets. A pocket for manuals, charts or personnel papers shall be furnished on the inside face of the door.

3.5.4.1.2 Pump cabinet lighting. The pump cabinet shall be furnished with not less than two 12-volt lamps, providing sufficient illumination for night operations.

* 3.5.4.2 Underslung cabinets. Underslung cabinets, constructed of steel of not less than 16 gage (0.0598 inch) thickness, shall be provided on both sides of the vehicle. Cabinets shall be approximately equal in size, and shall extend from the rear wheelhousing forward to the front of the tank. Cabinets shall be suspended from crossmembers or frame outriggers. Each cabinet shall be of full flanged, flush type, constructed to carry at least 700 pounds and of a size to house the

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specified equipment. Side cabinets shall have wooden trays suitable for stowing strainers, couplings and tools, and racks or shelves of the dimensions required to stow the discharge hose specified in 3.6.7. Doors shall be sidehung and equipped with continuous rod or hem type hinges with stainless steel hinge pins and stainless steel, bronze, nylon, or nylon insert sockets. Cabinets shall have provisions for flushing, draining, and for ventilating through fixed louvers. Doors shall have latches and fittings for locking with a padlock. Door handles shall be capable of operation by personnel wearing arctic gloves. Each door shall be furnished with a cam type lock with vertical drawbolts at the door header and door sill.

3.5.5 Pump. An auxiliary gasoline engine driven pump shall be furnished. The pump and auxiliary engine shall be installed on a base plate and bolted to crossmembers through the floor of the rear cabinet and shall be easily removable for maintenance and service. The pump and auxiliary engine shall be mounted to minimize transfer of vibration. Unions or pipe flanges shall be furnished at the pump for disassembly of the unit. Lifting eyes shall be furnished. Provisions shall be made for draining the pump to the ground.

3.5.5.1 Pump size. A 3-inch centrifugal pump shall be provided. The pump shall have a design capacity of 300 gallons per minute at a 45-foot head, continuous duty, and shall be capable of pumping and discharging sewage and waste containing particles up to 1-1/2 inches in diameter. An easily removable coverplate to facilitate cleaning the pump shall be furnished. A suction guard shall be provided on the suction port to preclude large objects from being drawn into the pump. Replaceable wear plates shall be furnished. The pump shall be provided with a check valve and shall be self-priming.

3.5.5.2 Auxiliary engine. An air-cooled gasoline engine shall be furnished to drive the pump. During normal operating conditions, the continuous-duty horsepower output required by all components and accessories driven by the engine shall not exceed 75 percent of the full throttle rating of the engine. The size and weight of the engine assembly shall be held to a minimum consistent with the performance requirements specified herein. Provisions shall be made for draining the crankcase from its installed position in the pump cabinet to ground without spillage on the cabinet floor.

3.5.5.2.1 Engine components. The engine shall be a four cycle type equipped with at least the following components:

(a) Distributor or magneto type ignition
(b) Self-starter, 12-volt

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(c) Resistor-type spark plugs
(d) Air cleaner
(e) Choke
(f) When specified (see 3.4.1.6), a power plant heater having a 300-watt immersion type heater with 170°F to 195°F thermostat shall be installed in the oil pan through any convenient opening
(g) Battery, 12-volt
(h) Alternator or generator, 12-volt

3.5.5.3 Electrical system. The auxiliary engine electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.33 and shall be vaporproof.

3.5.5.4 Fuel pump. A fuel pump shall be furnished on the engine. When lifting fuel against a head of 4 feet, the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line gasoline fuel filter shall be provided, either as a component of the fuel pump or between the fuel pump and the carburetor.

3.5.5.5 Governor. A variable-speed governor shall be furnished. The governor shall automatically control the engine speed at the selected rated speed at all pressures and flows required. A manually operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

3.5.5.6 Auxiliary engine fuel tank. An auxiliary engine fuel tank shall be furnished. The fuel tank shall have a capacity to provide fuel for at least 8 hours of full-load operation. The fuel tank cap shall be painted red and shall be attached to the tank by means of a corrosion-resistant chain. A sump shall be provided in the fuel tank suitable for collection and accumulation of water. A drain shall be located at the bottom of the sump for complete drainage of the fuel tank. The drain shall be located to preclude spillage of fuel on any area of the vehicle. The tank filler shall be provided with a protective screen and vent. A shut-off valve shall be provided in the fuel line to the engine adjacent to the fuel tank. A corrosion resistant plate shall be furnished and marked denoting the type of fuel to be used, and shall be located in a conspicuous location near the filler cap. The fuel tank and system shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67.

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* 3.5.5.7 Exhaust system. The auxiliary engine exhaust system shall be furnished with a spark arrester. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350. Means shall be provided to vent exhaust fumes away from the cab and the body.

3.5.5.8 Radio suppression. Radio suppression shall be in accordance with 3.4.2.7.

3.5.5.9 Control panel. A control panel shall be furnished inside the pump cabinet. The control panel shall consist of at least the following components, parts and accessories:

(a) Engine accelerator
(b) Palm button starter
(c) Key operated ignition switch
(d) Ammeter
(e) Oil gage
(f) Gasoline gage
(g) When body heaters are specified (see 3.8), gasoline heater and strip heater switches, with a red indicator light for each located on the control panel and on the vehicle dashboard
(h) Dash lights, with rotating shields and toggle switch, to illuminate the control panel.

All panel mounted components, accessories and parts shall be mounted so as to minimize vibration and fatigue. Grommets shall be employed where necessary. Panel shall be shock mounted so as to minimize transfer of vibration. Panel shall be weatherproof and waterproof. Each function shall be identified. A schematic on a corrosion-resistant plate depicting the function and source shall be furnished. The schematic shall be located on the inside of the rear cabinet door.

3.6 Piping. Piping shall be seamless steel extra strong pipe having a test pressure of not less than 1300 pounds per square inch (psi). Pipe shall be 3.500 inch outside diameter having a wall thickness of not less than 0.300 inch, and the outside diameter tolerance shall vary not more than plus or minus 1 percent.

3.6.1 Cleanout. A cleanout, not less than 3 inches in diameter, shall be furnished at the bottom rear end of the tank. The cleanout shall be provided with a 3-inch valve, nipple and brass cap with corrosion-resistant retaining chain.

3.6.2 Level indicator. A level indicator shall be provided to indicate the amount of fluid in the tank. The indicator shall be marked in increments of not more than 100 gallons. The indicator shall be located in the pump cabinet.

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3.6.3 Float valve. A float valve shall be furnished. The float valve shall be installed to stop the pump automatically when the tank has reached its rated capacity.

3.6.4 Tank outlet valve. A 3-inch diameter tank outlet valve of the flanged type shall be provided. The outlet valve, when closed, shall seal the tank, and shall be of the type permitting the complete removal of the valve body from outside the tank. The valve shall be located on the manifold in the pump cabinet.

3.6.5 Manifold. A manifold shall be furnished. The manifold shall be located in the pump cabinet. The manifold shall be furnished with 3-inch gate valves, and with the cam locking connections designed to lock to the hoses specified herein. The manifold shall provide the following functions:

(a) Fill the tank by suction through the pump and by pressure to the tank
(b) Pressure discharge of the tank contents by use of the pump
(c) Gravity discharge of the tank contents, bypassing the pump
(d) Fill the tank through the filler opening by utilizing the pump unit and hoses specified herein, from a non-pressurized source

* 3.6.6 Suction hose. Four 10-foot lengths of smooth-bore suction hose shall be furnished. Suction hose shall be in accordance with grade A, style A, Class I, size 3 of ZZ-H-561, and shall withstand handling and usage in ambient temperatures of plus 135°F to minus 40°F. Male and female couplings shall be furnished on each length of hose and shall be of the quick-disconnect, cam-locking type. Suction hose shall be stowed in the hose tubes specified in 3.5.3.

* 3.6.7 Discharge hose. Three lengths of single jacket discharge hose, each 25 feet in length and 3 inches in diameter, shall be furnished. The discharge hose shall be in accordance with type I, class A, size 3 of ZZ-H-451. The hose shall withstand handling and usage in ambient temperatures of plus 135°F to minus 40°F. Male and female couplings shall be furnished for each length of hose and shall be of the quick-disconnect, cam-locking type. The discharge hose shall be stowed on racks or shelves in the side cabinets specified in 3.5.4.2.

3.6.8 Strainer. A strainer shall be furnished for the suction hose. The strainer shall be easily demountable to permit cleaning and replacement.

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3.6.9 Special tools. Such tools as are normally required for operation of the equipment shall be furnished and stowed in the left side cabinet. Hold-down spring type clamps to secure all tools shall be provided.

3.7 Insulation. When specified (see 6.2), the tank, pump cabinet and all piping shall be completely insulated. Insulation shall provide a K factor of not more than 0.15 and a density of not less than 2.0 pounds per cubic foot. Insulation shall be asbestos-free, shall not support combustion or sustain flame, mold, rot or vermin, corrode metals, absorb odors, shall be essentially odorless, and shall not disintegrate under vibration or settle and leave voids.

3.7.1 Tank insulation. Insulation shall completely encircle the tank, including the tank ends. The tank insulation shall be covered with an outer jacket of type 304 stainless steel. The stainless steel jacket shall be welded on the outside and shall be waterproof. The minimum thickness of the outer jacket shall be 0.035 inch. The outer jacket shall completely enclose the tank, including under the tank and on the ends.

3.7.2 Insulated pump cabinet. The pump cabinet sides, ceiling, doors and floor shall be insulated. The cabinet shall have inner panels, completely covering the insulation.

3.7.3 Pipe insulation. All piping, the manifold and all fittings shall be insulated. Insulation shall be fiberglass pipe insulation, covered with a vapor barrier. The insulation and its covering shall be weatherproof and flame-resistant.

3.8 Body heaters. When specified (see 6.2), the pump cabinet and the tank shall have heaters as specified herein.

3.8.1 Heater for pump cabinet. A gasoline burning hot air heater shall be furnished for the rear pump cabinet. Fuel for the heater shall be supplied from the auxiliary engine fuel tank. The fuel system shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67. The capacity of the heater shall be not less than 60,000 British thermal units (Btu). The heater shall be capable of starting and operating at minus 45°F. The heater shall be thermostatically controlled and means shall be provided for automatic start and stop. The heater shall be installed in the cabinet on a base plate, bolted through the cabinet floor to crossmembers. The heater shall be provided with lifting eyes. The heater shall be readily removable for maintenance purposes. Means shall be provided to stop the heater automatically when the vehicle is in motion. Installation of the heater shall be in accordance with Federal Motor Carrier Safety Regulation 393.77.

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3.8.2 Tank heaters. The tank shall be provided with 110-volt, 60-cycle, single-phase strip heaters. Strip heaters shall maintain water temperature at not less than 35°F when the vehicle is in an ambient temperature of minus 45°F. Wiring shall be in conduit. An on-off switch shall be furnished, located on the control panel in the pump cabinet. A thermostat to sense the temperature of the liquid in the tank shall be provided, with thermostat controls located on the control panel in the pump compartment. Strip heaters shall be wired so that, in the event any circuit should fail, the other circuits continue to operate. Each circuit shall be provided with a means for checking if that circuit is operating.

3.8.3 Power supply receptacle. A 110-volt, 3-wire, single-phase, 60-cycle polarized, waterproof receptacle shall be furnished for power to the strip heaters. The receptacle shall be provided with a spring-loaded cover and shall be installed at the curbside rear of the pump cabinet. The receptacle shall be grounded.

3.8.4 Cable. A cable rated for 110-volt, 3-wire, single-phase, 60-cycle power shall be furnished. The cable shall be not less than 100 feet in length. The vehicle end of the cable shall mate with the receptacle specified in 3.8.3. The other end shall be provided with a polarized twist lock fastener and mating connector for installation by the using activity for connecting to a power source. The cable electrical capacity shall be adequate for full current draw with all strip heaters operating. The cable insulation shall be of neoprene rubber and of sufficient dielectric capacity to meet the imposed dielectric loads. The cable shall be stowed in a cabinet.

* 3.9 Body mounting. The body shall be secured with U-bolts, twin studs, or by brackets, and shall include a wood breaker strip. All but the rearmost pair of U-bolts, twin studs or brackets shall be spring loaded to permit vertical movement between the chassis and the body.

3.9.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than three U-bolts or twin studs, per side, each having 9/16-inch diameter with 5/8-inch minimum thread diameter. Tie-plates shall be at least 1/2 inch thick. Vehicle chassis frame shall be braced, using wood blocks at each mounting point unless the mounting point is located at a full depth frame crossmember. Blocks shall incorporate a keeper strap or groove for mounting bolt, and shall be of a width and thickness to assure retention. Two tie-back straps shall be provided, one on each side of the rear portion of the body subframe, to maintain body alinement on the vehicle chassis. Forward body mounting bolts shall be located to the rear of the tapered portion of breaker strips (see 3.7.3).

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3.9.2 Brackets. When brackets are used, they shall be bolted to the web of the chassis frame rails. The body mounting brackets shall provide means of drawing down the body on the chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure mounting brackets to chassis frame rails, they must be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components.

3.9.3 Breaker strips. A hardwood or dense southern yellow pine breaker strip not less than 1-1/16 inches finished thickness, shall be installed between longitudinal sills and vehicle chassis frame. Breaker strips shall have a taper of 1 inch in 18 inches at the forward end.

3.10 Servicing and adjusting. Prior to acceptance of each vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights, adjustment of engine, adjustment of electrical and brake systems, charging of batteries, alinement of front wheels, inflation of all tires, complete lubrication of chassis, engines, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

3.11 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts, and castings shall be commensurate with sound manufacturing standards for similar equipment. The vehicles shall be free from all burrs, sharp edges, die, tool, and metal handling marks, rust, scale, and cracks. All metal and plastic materials used shall be free from delamination defects, and shall be prime material. Improper manufacturing or assembling practices, defective components or parts and assemblies, which have been modified to overcome deficiencies, shall not be furnished.

* 3.12 Welding procedures. Body welding procedures shall be prepared by the contractor performing the welding and shall be in accordance with a nationally approved code. Welding procedures shall be identified and made available to the Government representative for review. (see 4.2.1).

* 3.12.1 Welding workmanship. All body welds shall be free from slag, flux, cracks, porosity and any other discontinuity detrimental to the strength and soundness of the weld. All body welds shall have a

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thorough fusion between adjacent layers of weld metal and between weld metal and base metal. All but weld profiles shall be slightly covex in contour and, when specified on detail drawings, shall be full penetration. (see 4.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

* 4.2.1 In-process inspection. The first production tank body being produced under the contract shall receive an in-process inspection by the Government prior to prime painting. The inspection shall be to evaluate welding workmanship, procedures and weld quality. (see 3.12 and 3.12.1).

4.3 First vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the

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direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on the front and rear axles. The imposed loading on front and rear axles shall be computed by the contractor and verified by the Government using the curb weight, the operator weight at 175 pounds, and the payload (see 3.2.5) to provide the specified GVW. Calculated imposed loads on front and rear axles shall be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet contractual requirements.

4.3.2 Road test. The first production vehicle shall be examined and road tested by the contractor to assure that the vehicle will operate in accordance with contractual requirements. The vehicle, loaded with payload and with all equipment specified herein, shall be driven at speeds up to at least 50 mph, for a minimum distance of 50 miles. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turns. For each 10 miles of travel, at least one sudden stop shall be accomplished. Upon completion of the road test all equipment shall be thoroughly examined. The equipment shall show no evidence of damage; all cabinet and cab doors shall properly open and shut with no trace of mud inside; and there shall be no evidence of racking of the body. The vehicle shall be tested in accordance with 4.3.4 after the road test.

* 4.3.3 Truck body treatment and painting. The certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be presented to Government representatives for examination and approval.

4.3.4 Leakage test. The completely assembled tank body, including all piping, shall be tested for leaks at its design pressure in accordance with DoT Safety Regulations Governing Transportation of Hazardous Materials, Section 178.341, Specification MC306. In addition, all piping and hoses shall be tested for leaks at the maximum pressure they can encounter during use of the vehicle-mounted pump.

4.3.5 Pump test. Vehicle pumping equipment shall be tested to determine performance requirements specified in 3.5.5.1.

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* 4.3.6 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data), if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

4.3.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.8 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle, shall be rejected until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall as a minimum assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. All vehicles under contract shall be tested for leaks as specified in 4.3.4. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.10 has been accomplished.

5. PACKAGING

5.1 Vehicle processing. Vehicles shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government in collecting and disposing of sewage.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Class, and model required (see 1.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).

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(d) Concealed marking, if required (see 3.1.1.1).
(e) Rustproofing, if required (see 3.1.1.2).
(f) Tropical rustproofing, if required (see 3.1.1.2).
(g) Towing devices on the rear of vehicle, if required (see 3.1.1.6).
(h) Thermostatically controlled radiator shutters, if required (see 3.1.1.9).
(i) Brushguard, if required (see 3.1.1.10).
(j) Gasoline engine, if required (see 3.4.1.2).
(k) Power plant heaters and fuel warmer, if required (see 3.4.1.6)
(l) If a battery heater is not required (see 3.4.1.6).
(m) Trailer lighting cable, if required (see 3.4.2.5).
(n) Two-speed transfer case, if required (see 3.4.5.1).
(o) Automatic or semiautomatic transmission, if required (see 3.4.5.3).
(p) Traction control, if required (see 3.4.9.2).
(q) Wide base type tires and wheels, if required (see 3.4.10).
(r) Disc type wheels, if required (see 3.4.10).
(s) Bias ply tires, if required (see 3.4.10.1).
(t) A carrier for a spare tire assembly, if required (see 3.4.10.3).
(u) Spare wheel or rim, if required (see 3.4.10.4).
(v) Spare tire assembly, if required (see 3.4.10.5).
(w) Air dryer, if required (see 3.4.11.1 (k)).
(x) Brake controls for use from a towing vehicle, if required (see 3.4.11.3).
(y) Individual adjustable driver's and individual passenger seat, if required (see 3.4.12).
(z) Tools, if required (see 3.4.16.1).
(aa) Air horn, in addition, if required (see 3.4.21).
(ab) Backup alarm, if required (see 3.4.23).
(ac) Air conditioning, if required (see 3.4.24).
(ad) Front mounted winch, if required (see 3.4.25).
(ae) Insulation, if required (see 3.7).
(af) Body heaters, if required (see 3.8).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure Form and computation for low speed requirement for each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet item 1 should include vehicle model number, wheelbase, and engine model number. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still,

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dry, air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required
Table 9	- Road Factor. 0.0.

6.4 Identification of changes. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, regardless of the marginal notations and relationship to the last previous issue.

Custodian:
Army - AT

Preparing Activity:
Army - AT
Project No. 2320-A274

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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